

SURVEY-WEIGHTED ORDINAL CAUSAL DISCOVERY: APPLICATION TO VETERAN SOCIAL NEEDS AND HEALTH INDICATORS

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Health-related social needs ("social needs") – the midstream manifestations of social determinants or drivers of health – are highly intercorrelated, and disentangling their direct from indirect influences on health requires knowledge of the underlying causal structure. Standard associational methods cannot recover this structure, and causal discovery algorithms offer a data-driven alternative. However, existing methodology, designed for independent and identically distributed ordinal data, yields biased results when applied to ordinal data collected from surveys. To overcome this limitation, we introduce the survey-weighted ordinal Bayesian network, or *swa-oBn*. We provide technical guarantees that *swa-oBn* consistently recovers the true population-level causal graph in large samples, even under complex survey designs. We apply *swa-oBn* to national Veterans Health Administration (VHA) Survey of Healthcare Experiences of Patients data to map the causal architecture of Veterans' social needs and self-reported mental health and well-being. We demonstrate that ignoring the complex survey design recovers a structurally different graph in which the combined grouping of social isolation, loneliness, and discrimination is a root cause of material deprivation and health outcomes. However, by explicitly modeling the sampling design, *swa-oBn* recovers the following structural hierarchy: unmet basic needs are the root cause, with social isolation, loneliness, and discrimination acting as a downstream mediator to poor mental health instead. This distinction has direct intervention consequences: targeting isolation, loneliness, and discrimination without also addressing material deprivation may produce limited health gains. Complementary simulations confirm that ignoring complex survey designs produces biased causal graphs, with qualitatively different and potentially misleading policy conclusions.

1. Introduction. Health-related social needs (hereafter, social needs) – the midstream manifestations of social determinants or drivers of health (Castrucci and Auerbach, 2019) – such as financial instability, housing insecurity, and social isolation are highly intercorrelated in population surveys, creating a fundamental challenge for effective intervention design. Standard regression analyses document associations between unmet social needs and poor health outcomes (Hausmann et al., 2023), but they cannot differentiate upstream root causes from downstream mediators (Pearl, 2009). This distinction is critical for both program and policy development. For example, if *underemployment* or *limited education* are root causes that cascade into *unmet basic needs* (e.g., housing instability, food insecurity, and inability to

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pay for basics) and *psychosocial challenges* (e.g., social isolation, loneliness, and discrimination) as well as poor mental health, then interventions targeting employment and education will have broad downstream benefits. However, if the causal ordering is different, so are the optimal program and policy foci and resource allocation (Athey, 2017). Social needs — and health-related constructs more broadly — are typically measured on ordinal scales (Streiner, Norman and Cairney, 2024), making the recovery of causal structure from such data both a practical necessity and a methodological problem.

Causal discovery algorithms offer a data-driven methodology for inferring directed causal structure from observational data (Glymour, Zhang and Spirtes, 2019). However, applying causal discovery to survey data introduces a methodological problem that, to our knowledge, has not been addressed in the literature: complex surveys produce non-independent and identically distributed (non-i.i.d.) data, and ignoring the survey design yields distorted or biased underlying causal relationships, often represented using Directed Acyclic Graphs (DAGs).

To illustrate this problem concretely, consider a simple population causal chain $X \rightarrow Y \rightarrow Z$, where X , Y , and Z are ordinal variables representing basic financial need, access to resources, and well-being. In the population, X and Z are conditionally independent given Y . Now suppose a survey over-samples individuals with both high X (severe need) and low Z (poor well-being). The resulting sample over-represents the joint tail of (X, Z) , inducing a spurious marginal dependence between X and Z that persists even after conditioning on Y . An unweighted causal discovery algorithm, confronted with this distorted distribution, will infer a direct edge $X \rightarrow Z$ (or $Z \rightarrow X$) that does not exist in the population, producing a biased causal graph that reflects the *sampling mechanism* rather than the *data-generating process*. This bias is systematic, worsens with stronger oversampling, and does not vanish as the sample size increases.

The survey design problem is thus the first obstacle. The natural tool for causal discovery with observational data is the Bayesian network, which represents the causal structure as a DAG (Pearl, 2009). But for ordinal variables, standard multinomial Bayesian networks are non-identifiable: multiple DAGs encoding the same conditional independence relationships produce identical likelihoods (e.g., Markov equivalence), so edge directions cannot be recovered even with infinite data (Chickering, 2002). Ni and Mallick (2022) resolved this identifiability problem for ordinal data by introducing the ordinal Bayesian network (oBn). They proved that the ordinal parameterization breaks Markov equivalence when all variables have three or more levels, enabling full DAG identifiability from observational data. Ni, Chen and Wang (2025) extended this to a bootstrapped multivariate algorithm with a BIC-based score and demonstrated its utility in a study of obsessive-compulsive disorder and depression. To our knowledge no other approach has been developed specifically for ordinal data with provable full DAG identifiability.

However, the oBn framework assumes i.i.d. data, does not support exogenous covariate adjustment, and uses a single-BIC score that introduces a finite-sample orientation bias when ordinal variables have heterogeneous numbers of levels. This paper addresses these gaps by introducing the *survey-weighted, covariate-adjusted ordinal Bayesian network*, or swa-oBn . The method makes four contributions over the existing oBn framework:

1. *Survey weighting.* We propose a pseudo-BIC scoring framework, enabling design-consistent structure learning from complex survey data.
2. *Covariate adjustment.* Measured demographic confounders are absorbed into every node-wise regression without expanding the DAG, reducing confounding bias while preserving the ordinal identifiability guarantee.
3. *Scoring variants for heterogeneous ordinal scales.* The scoring proposed by Ni and Mallick (2022); Ni, Chen and Wang (2025), which we call the single-BIC routine, scores all moves — additions, deletions, and reversals — on the penalized pseudo-likelihood.

We introduce and justify the use of *split-BIC*, which scores reversals on the unpenalized pseudo-likelihood while retaining the penalized pseudo-BIC for additions and deletions. Both are large-sample consistent, and split-BIC outperforms single-BIC at the sample sizes tested (Section 7).

4. *Structural constraints.* Domain knowledge is encoded via blacklisted and whitelisted edges, restricting the search space to scientifically plausible graphs.

We prove that the `swa-oBn` consistently recovers the true population-level DAG in large samples under a joint superpopulation-and-design asymptotic framework.

We apply `swa-oBn` to national data from the “Your Recent Visit” Veterans Health Administration (VHA) Survey of Healthcare Experiences of Patients (SHEP) to map the causal architecture linking self-reported social needs to health indicators (i.e., self-reported mental health and well-being) among U.S. Veterans. Prior associational analyses have documented substantial variation in need prevalence across demographic groups (Frank et al., 2025) and shown that standard risk screening under-detects needs in some subgroups (Russell et al., 2026), but have not addressed the structural question: which needs are upstream root causes, and which are downstream mediators? The answer depends on whether the survey design is modeled. *An unweighted oBn analysis recovers a graph in which social isolation, loneliness, and discrimination act as a root cause, with direct edges to unmet basic needs and to health indicators — implying that interventions should target psychosocial behavior first. The swa-oBn analysis reverses this: unmet basic needs emerge as the upstream root cause, while social isolation, loneliness, and discrimination function as the downstream mediator to poor mental health. The two graphs point to opposite intervention priorities, and the difference is driven entirely by the survey design — not by the data, the variables, or the algorithm.*

The remainder of this paper is organized as follows. Section 2 introduces the SHEP data and the scientific questions motivating the analysis. Section 3 reviews ordinal Bayesian networks. Section 4 develops the `swa-oBn` methodology. Section 5 establishes theoretical properties. Section 6 reports the application to VHA data. Section 7 presents simulation studies. Section 8 concludes with a discussion and implications of our findings. An R implementation of the proposed methodology, the `swa-oBn` package, is available with full documentation online.

2. Data and Scientific Questions.

2.1. The Veterans Health Administration and social needs. The VHA is the largest integrated healthcare system in the United States, providing care to over nine million Veterans enrolled in VHA healthcare programs across the country (U.S. Department of Veterans Affairs, 2026). Understanding and addressing social needs among this population has become a major policy objective (Daniel, Bornstein and Kane, 2021). These efforts also align with organizational priorities and The Joint Commission accreditation standards (List et al., 2024). The existing literature on social needs in the Veteran population has extensively documented that unmet social needs are associated with poor health outcomes (Braveman, Egerter and Williams, 2011; Duan-Porter et al., 2018; Hatef et al., 2019). Recent analyses of the VHA patient population have revealed substantial variation in social need prevalence: compared with White male Veterans, Black male Veterans report elevated needs in 11 of 13 social need domains, and need for support managing experiences of discrimination is elevated across all non-White-male Veteran subgroups (i.e., White, Black, and Hispanic women and Black and Hispanic men: Frank et al. (2025)). These findings confirm that social needs are pervasive and heterogeneous within the Veteran population. Thus far, analyses investigating variation of social needs have been associational. Regression models report which social needs are correlated with specific health indicators; however, they do not reveal the causal ordering among

said needs, leaving organizational leaders and policymakers without the evidence required to develop and prioritize interventions and policy decisions effectively. This paper aims to provide said evidence and inform the following discussions of scientific and practical importance that standard associational methods cannot answer:

1. Which social needs are upstream root causes of other needs, and which are downstream consequences?
2. Among the needs causally linked to poor health indicators, which function as *distal* causes (i.e., operating on outcomes indirectly through mediating needs) and which function as *proximal* causes that transmit upstream effects directly to health?

2.2. Survey design and data. We used data collected through the VHA "Your Recent Visit" survey, which is administered as part of VHA's SHEP program. Patterned after the Consumer Assessment of Healthcare Providers and Systems (CAHPS) surveys ([Agency for Healthcare Research and Quality, 2024](#)), SHEP is administered to a national stratified random sample of VHA primary care patients each month. This study includes survey responses from a stratified random sample of Veterans who presented for primary care between January and February 2023. As described elsewhere ([Frank et al., 2025](#)), to ensure adequate statistical power for intersectional comparisons, the design deliberately oversampled Black and Hispanic male Veterans and Black, Hispanic, and White female Veterans, producing six race-ethnicity-sex strata with approximately 900 respondents each. The analytic sample comprised $n = 5,396$ respondents with complete data across all social need domains, outcome variables, and covariates used in the analysis. The SHEP non-response weights adjust for unit-level survey non-response given the sampling frame but are not recalibrated for item-level missingness producing the complete-case subset. Survey weights vary substantially across respondents: the ratio of the largest to smallest weight is approximately 980:1, with a coefficient of variation of 1.70. Kish's effective sample size ([Kish, 1965](#)) is $n_{\text{eff}} = 1,387$, yielding a design effect of 3.89. This effective sample size governs the penalty term in the pseudo-BIC (see Section 4.2) and reflects a nearly fourfold reduction in effective information relative to a simple random sample of the same size.

2.3. Need variable construction. The survey assessed need for support in the past six months across 13 social needs using a three-level ordinal scale: "no support needed" (level 1), "needed support and got it" (level 2), and "needed support but did not get it" (level 3). The analysis retained the full ordinal scale: the identifiability result of [Ni and Mallick \(2022\)](#) that underlies our framework requires ordinal variables with three or more levels.

A potential challenge in applying causal discovery to survey instruments is the high collinearity of item-level data, particularly when items target overlapping constructs. To define conceptually distinct nodes for the oBn , we analyzed the item-level polychoric correlation structure. Items exhibiting polychoric correlations $\rho \geq 0.75$ were aggregated into composite ordinal domains by taking the highest level of need across the constituent items on the three-level scale (no support needed < needed support and got it < needed support but did not get it), thus preserving the original three-level ordinal scale. This yielded three composite nodes: *basic needs* (paying for basics like food, housing, medical care, and heating; paying for only food; getting or maintaining housing; $\rho = 0.85\text{--}0.95$), *work/education* (finding or keeping work; getting additional education or job training; $\rho = 0.82$), and *isolation/loneliness/discrimination* (feeling socially isolated; feeling lonely; managing experiences of discrimination; $\rho = 0.75\text{--}0.96$). Five items falling below the polychoric correlation threshold of $\rho \geq 0.75$ were retained as standalone nodes: *adult caregiving* (for self or others), *childcare*, *transport* (transportation for basic needs like medical care or grocery shopping),

internet (access at home), and *legal assistance* (see full social needs survey items in Supplementary Material). The downstream health indicators were captured by two additional ordinal nodes: well-being (four levels) and mental health (five levels). The well-being node was derived from three questions asking respondents to estimate, on a 0 – 100% scale, the percentage of time during the past four weeks that (i) their physical health was good, (ii) their emotional health was good, and (iii) they were able to perform their usual social activities; these three percentages were averaged into a composite well-being score and discretized using fixed cutpoints at 25%, 50%, and 75% (i.e., $[0, 25]$, $(25, 50]$, $(50, 75]$, $(75, 100]$), yielding four ordered levels in which level 1 corresponds to the worst well-being and level 4 to the best. We use fixed cutpoints rather than data-driven quartiles to preserve the absolute interpretation of the underlying percentage scale and to keep the discretization invariant under reweighting and resampling. The mental health node was taken directly from a single question asking respondents to rate their mental health, with five response options (poor < fair < good < very good < excellent) coded ordinally.

The complete node set consists of $q = 10$ ordinal variables with 3 – 5 levels each. The polychoric correlation matrix and clustering are presented in the Supplementary Material. The polychoric correlation threshold $\rho \geq 0.75$ was chosen in collaboration with domain experts (co-authors DF, LR, and LH) representing a range of professions to balance construct validity with network interpretability. We treated this as a deliberate methodological trade-off, accepting an approximate working model in exchange for an interpretable graph.

2.4. Covariate adjustment and structural constraints. To mitigate confounding by demographic characteristics, our analysis adjusts for age (categorized into five bands: 18 – 44, 45 – 54, 55 – 64, 65 – 74, and ≥ 75 years) and a six-category combined race-ethnicity-sex indicator (White males, Black males, Hispanic males, White females, Black females, and Hispanic females). These variables are not treated as nodes in the DAG; instead, they form an exogenous covariate matrix $\mathbf{Z}$, ensuring the discovered edges represent relationships adjusted for demographic confounding. A discussion of what $\mathbf{Z}$ does and does not absorb is deferred to Section 8.

Finally, we integrated domain knowledge via a structural blacklist. All directed edges originating from the two health indicator nodes (mental health and well-being) and pointing toward any of the eight social need nodes are forbidden. This encodes the consensus in the social needs literature that unmet social needs predispose individuals to poor mental and physical health, rather than the reverse (Commission on Social Determinants of Health, 2008; Adepoju et al., 2022). We acknowledge that reverse pathways are plausible (e.g., depression can precipitate job loss or social withdrawal) and that cross-sectional self-reported data cannot resolve such bidirectional dynamics; we therefore treat the blacklist as a deliberate scientific decision rather than an empirical claim. Within the paradigm that needs are upstream of outcomes, our scientific interest lies in the causal network *among* the needs themselves, since this network identifies which upstream needs are the highest-leverage targets for intervention. All relationships among the social need variables themselves, and the relationship between the two health indicators, were left entirely unconstrained.

3. Background: Ordinal Bayesian Networks. A Bayesian network (BN) represents the joint distribution of random variables $\mathbf{X} = (X_1, \dots, X_q)$ through a directed acyclic graph (DAG) $\mathcal{G} = (V, E)$ and an associated factorization $P(\mathbf{X} | \mathcal{G}) = \prod_{j=1}^q P(X_j | \mathbf{X}_{\text{pa}(j)})$, where $\text{pa}(j)$ denotes the parent set of node j in $\mathcal{G}$ (Pearl, 2009). Under the standard assumptions of causal sufficiency and faithfulness (see Section 5.1) the edges of $\mathcal{G}$ admit a causal interpretation. For categorical variables, each conditional $P(X_j | \mathbf{X}_{\text{pa}(j)})$ may be specified by a multinomial distribution. The multinomial parameterization is identifiable only up to a

Markov equivalence class (Heckerman, Meek and Cooper, 1997) and, when the variables are ordinal, discards informative structure. Ni and Mallick (2022) proposed the ordinal Bayesian network (oBn), which replaces multinomial conditionals with cumulative link regressions. For an ordinal variable X_j with L_j levels and parent set $\text{pa}(j)$, the oBn specifies

$$(1) \quad \Pr(X_j \leq \ell \mid \mathbf{X}_{\text{pa}(j)}) = \Phi \left(\gamma_{j\ell} - \sum_{k \in \text{pa}(j)} \beta_{jk} X_k - \alpha_j \right), \quad \ell = 1, \dots, L_j,$$

where Φ is the standard normal cumulative distribution function (any strictly increasing link F may be used; we adopt the probit function for consistency with Ni and Mallick (2022)), $\gamma_{j1}, \dots, \gamma_{j,L_j-1}$ are ordered cutpoints with $\gamma_{j1} = 0$ for identifiability, α_j is an intercept, and β_{jk} captures the effect of parent X_k on X_j . When a parent X_k enters as a predictor with L_k levels, separate coefficients $\beta_{jk1}, \dots, \beta_{jk,L_k-1}$ replace the scalar β_{jk} , with $\beta_{jk1} = 0$ as a reference constraint. The total count of free parameters under DAG $\mathcal{G}$ is

$$(2) \quad K_{\mathcal{G}} = \sum_{j=1}^q \left(L_j - 1 + \sum_{k \in \text{pa}(j)} (L_k - 1) \right),$$

and enters the BIC penalty (see Section 4.1).

The central property of the oBn is identifiability. Ni and Mallick (2022) proved that if $X_j \rightarrow X_k$ is the true direction and both variables have three or more levels, the reverse model $X_k \rightarrow X_j$ cannot reproduce the same joint distribution except on a Lebesgue-null subset of the parameter space. Ni, Chen and Wang (2025) extended this to the multivariate setting. The implication is that the true DAG is distinguishable from all Markov-equivalent alternatives, making score-based structure learning viable for ordinal data. However, the oBn framework assumes i.i.d. data and does not accommodate survey weights or exogenous covariates. The next section develops swa-oBn with extensions required to address these limitations.

4. Survey-Weighted Ordinal Causal Discovery. We now develop the survey-weighted, covariate-adjusted ordinal Bayesian network (swa-oBn), extending the oBn of Ni and Mallick (2022); Ni, Chen and Wang (2025) to complex survey data with measured confounders and domain knowledge. The method has four components: a node-wise scoring framework with two variants, single-BIC and split-BIC, in 4.1; survey weighting in 4.2; covariate adjustment in 4.3; and bootstrap uncertainty quantification in 4.4. Algorithm pseudocode is in the Supplementary Material.

4.1. Node-wise scoring: single-BIC and split-BIC. Let $\mathbf{x} = (x_1, \dots, x_n)$ denote n realizations of $\mathbf{X}$. Under the oBn factorization with cumulative probit conditionals (1), the log-likelihood decomposes node-wise,

$$(3) \quad \ell(\boldsymbol{\theta}; \mathcal{G}) = \sum_{j=1}^q \ell_j(\boldsymbol{\theta}_j; \text{pa}(j)), \quad \ell_j(\boldsymbol{\theta}_j; \text{pa}(j)) = \sum_{i=1}^n \log p(x_{ij} \mid \mathbf{x}_{i,\text{pa}(j)}, \boldsymbol{\theta}_j),$$

with $\boldsymbol{\theta}_j = (\gamma_j, \boldsymbol{\beta}_j, \alpha_j)$. Ni, Chen and Wang (2025) scored candidate graphs by

$$(4) \quad \text{BIC}(\mathcal{G}) = -2\ell(\hat{\boldsymbol{\theta}}; \mathcal{G}) + K_{\mathcal{G}} \log n,$$

where $K_{\mathcal{G}}$ is given by (2), and select the DAG minimizing this score.

Exhaustive enumeration is infeasible: the number of DAGs on q nodes is super-exponential in q , and every graph requires fitting a node-wise cumulative link regression for each parent set. Score-based structure learning therefore proceeds by greedy hill-climbing (Chickering,

2002), which navigates the DAG space through local moves. From an initial graph, each iteration considers three move types subject to acyclicity: additions and deletions, which alter the skeleton, and reversals, which flip the direction of an existing edge. The node-wise decomposition (3) makes this efficient: a single-edge modification changes the parent set of at most two nodes, so only those node-wise scores need recomputation. The search terminates at a local optimum; we use R random restarts and return the solution with the lowest penalized score.

Edge reversals are the consequential case for scoring: they navigate within Markov equivalence classes, where ordinal identifiability (Ni and Mallick, 2022) is the only mechanism distinguishing $k \rightarrow j$ from $j \rightarrow k$. When all variables share a common number of levels L , the parameter counts of $k \rightarrow j$ and $j \rightarrow k$ coincide and the BIC penalties cancel within any equivalence class; orientation is governed entirely by the likelihood. When ordinal scales are heterogeneous, the regression of X_j on X_k contributes $L_k - 1$ slopes while the reverse contributes $L_j - 1$, so Markov-equivalent DAGs differing only in edge direction receive different BIC penalties of size $(L_k - L_j) \log n$. The single-BIC score of Ni and Mallick (2022); Ni, Chen and Wang (2025), which applies (4) uniformly to all move types, inherits this differential and biases reversal decisions toward making the lower- L variable the parent.

We propose *split-BIC* scoring, which retains (4) for additions and deletions but evaluates reversals on the unpenalized $-2 \ell(\boldsymbol{\theta}; \mathcal{G})$. This removes the penalty differential and reduces within-class comparisons to a likelihood ratio: by generic ordinal identifiability, the correctly oriented edge attains a strictly higher likelihood for almost all parameter values, regardless of L . Both variants are large-sample consistent (see Section 5); empirically, split-BIC outperforms single-BIC at the sample sizes tested (see Section 7).

4.2. Survey weighting. When inclusion probabilities are associated with the variables under study, the sampling mechanism is informative (Pfeffermann, 1993) and parameter estimates obtained by treating the sample as i.i.d. can be inconsistent for the corresponding population quantities. The graph learned from an unweighted analysis may then reflect the sampling design rather than the population-generating causal structure.

The standard correction is pseudo-maximum likelihood estimation (Binder, 1983). Let S denote a probability sample of size n drawn from a finite population of size N , with first-order inclusion probabilities $\pi_i = \Pr(i \in S)$ and survey weights $w_i = 1/\pi_i$. We replace the log-likelihood (3) with the pseudo-log-likelihood

$$(5) \quad \ell^w(\boldsymbol{\theta}; \mathcal{G}) = \sum_{i \in S} w_i \log p(\mathbf{x}_i | \mathcal{G}, \boldsymbol{\theta}).$$

The node-wise factorization of (3) is preserved. The pseudo-BIC replacing (4) is

$$(6) \quad \text{pBIC}(\mathcal{G}) = -2 \ell^w(\hat{\boldsymbol{\theta}}^w; \mathcal{G}) (n_{\text{eff}}/n) + K_{\mathcal{G}} \log(n_{\text{eff}}),$$

where $\hat{\boldsymbol{\theta}}^w$ is the pseudo-maximum likelihood estimate and $n_{\text{eff}} = (\sum_{i \in S} w_i)^2 / \sum_{i \in S} w_i^2$ is Kish's effective sample size (Kish, 1965). From this point forward we adopt the convention that the weights in (5) are renormalized to sum to n , so that ℓ^w is on the sample scale; the factor n_{eff}/n in (6) then rescales it to the effective-information scale, matching the penalty scale (Lumley, 2004). When all weights are equal, $n_{\text{eff}} = n$ and (6) reduces to (4). The theory in Section 5.2 uses the equivalent raw-weight formulation; see Remark 3.

The two scoring variants described in Section 4.1 take their survey-weighted forms by replacing (4) with (6) for additions and deletions. For reversal moves, single-BIC retains the pseudo-BIC; split-BIC drops the penalty and uses the rescaled pseudo-log-likelihood $-2 \ell^w(\hat{\boldsymbol{\theta}}^w; \mathcal{G}) (n_{\text{eff}}/n)$.

4.3. *Covariate adjustment.* Measured covariates $\mathbf{Z} = (Z_1, \dots, Z_r) \in \mathbb{R}^r$ may confound relationships among $\mathbf{X}$ but are not themselves objects of causal discovery. We absorb their influence by extending (1) to include $\mathbf{Z}$ as an additive predictor at every node:

$$(7) \quad \Pr(X_j \leq \ell \mid \mathbf{X}_{\text{pa}(j)}, \mathbf{Z}) = \Phi \left(\gamma_{j\ell} - \sum_{k \in \text{pa}(j)} \beta_{jk} X_k - \mathbf{Z}' \boldsymbol{\delta}_j - \alpha_j \right), \quad \ell = 1, \dots, L_j,$$

where $\boldsymbol{\delta}_j$ is node-specific. Crucially, $\mathbf{Z}$ does not appear as a node in $\mathcal{G}$: the search adds, deletes, and reverses edges only among $\mathbf{X}$, so the search space remains $O(q^2)$ rather than $O((q+r)^2)$. The pseudo-BIC and scoring framework of Sections 4.1–4.2 apply unchanged with $\boldsymbol{\delta}_j$ absorbed into the parameter vector. Under this extension the causal sufficiency assumption applies to the enlarged variable set: all common causes of any pair (X_j, X_k) must lie in $\mathbf{X} \cup \mathbf{Z}$ rather than $\mathbf{X}$ alone. Identifiability is preserved: for each fixed $\mathbf{Z}$, model (7) reduces to an oBn with shifted intercepts, so Ni and Mallick (2022) continues to hold (see Proposition 1 in Section 5).

REMARK 1 (Structural constraints). A blacklist $\mathcal{B}$ of forbidden directed edges and a whitelist $\mathcal{W}$ of required directed edges encode domain knowledge. Blacklisted edges are never proposed and reversals creating one are skipped; whitelisted edges are placed in the initial graph and locked.

4.4. *Bootstrap uncertainty quantification.* A single hill-climbing run produces a point-estimate DAG $\hat{\mathcal{G}}$. To quantify structural uncertainty we use the nonparametric bootstrap. For each replicate $b = 1, \dots, B$, we resample n tuples $(\mathbf{x}_i, w_i, z_i)$ with replacement, recompute n_{eff} from the resampled weights, and re-run the search under the chosen scoring variant to obtain $\hat{\mathcal{G}}^{(b)}$. The edge inclusion probability for $j \rightarrow k$ is $\hat{p}_{jk} = B^{-1} \sum_{b=1}^B \mathbf{1}(j \rightarrow k \in \hat{\mathcal{G}}^{(b)})$, and a consensus graph retains edges with $\hat{p}_{jk} \geq 0.50$. This adapts the procedure of Ni, Chen and Wang (2025) to the survey-weighted setting; see pseudocode in Supplementary Material.

5. Theoretical Properties. We establish the large-sample properties of the swa-oBn under a joint asymptotic framework that accounts for two sources of randomness: the super-population ξ generating the finite population, and the sampling design selecting units from it. Let $U_N = \{1, \dots, N\}$ denote a finite population of size N , generated as N independent draws from ξ , with true causal DAG $\mathcal{G}^*$ and parameters $\boldsymbol{\theta}^*$. From U_N , a probability sample S_n of size $n = n(N)$ is drawn according to a sampling design $p_N(S)$ with known first-order inclusion probabilities $\pi_i = \Pr(i \in S_n)$ and survey weights $w_i = 1/\pi_i$. We require $N \rightarrow \infty$ and $n \rightarrow \infty$ with sampling fraction $n/N \rightarrow f \in [0, 1]$. All convergence statements are under the joint $(\xi \times p_N)$ -probability. Proofs are given in the Supplementary Material.

We denote $\boldsymbol{\theta}_{\mathcal{G}}^* = \arg \max_{\boldsymbol{\theta}} Q_{\xi}(\boldsymbol{\theta} \mid \mathcal{G})$; when $\mathcal{G} = \mathcal{G}^*$, this coincides with the true data-generating parameter, which we denote by $\boldsymbol{\theta}^*$. For a mis-specified DAG $\mathcal{G}' \neq \mathcal{G}^*$, we use $\boldsymbol{\theta}_{\mathcal{G}'}^{\dagger} \equiv \boldsymbol{\theta}_{\mathcal{G}'}^*$ to emphasize its role as the KL-projection rather than a truth-level parameter. The conditional (covariate-pointwise) KL-projection is denoted $\boldsymbol{\theta}^{\dagger}(\mathbf{z}) = \arg \max_{\boldsymbol{\theta}} \sum_{\mathbf{x}} P(\mathbf{x} \mid \mathcal{G}^*, \boldsymbol{\theta}^*, \mathbf{z}) \log P(\mathbf{x} \mid \mathcal{G}', \boldsymbol{\theta}, \mathbf{z})$, which in general differs from the unconditional $\boldsymbol{\theta}_{\mathcal{G}'}^{\dagger}$.

5.1. *Assumptions.* We organize the assumptions into four groups: causal assumptions (1–3), parametric assumptions (4–6), survey design assumptions (7–9), and a regularity assumption (10).

ASSUMPTION 1 (Causal Sufficiency). All common causes of any pair (X_j, X_k) in $\mathbf{X} = (X_1, \dots, X_q)$ are included in the measured variables $\mathbf{X} \cup \mathbf{Z}$.

ASSUMPTION 2 (Causal Markov Property). *The superpopulation distribution $P_\xi(\mathbf{X})$ factorizes with respect to the true causal DAG $\mathcal{G}^*$: $P_\xi(\mathbf{X}) = \prod_{j=1}^q P_\xi(X_j | \mathbf{X}_{\text{pa}(j)})$.*

ASSUMPTION 3 (Faithfulness). *Every conditional independence in P_ξ is entailed by the Markov property of $\mathcal{G}^*$: $X_A \perp\!\!\!\perp X_B | X_C$ in P_ξ if and only if A and B are d -separated by C in $\mathcal{G}^*$.*

ASSUMPTION 4 (Ordinal Regression Specification). *Each conditional distribution under the true DAG $\mathcal{G}^*$ follows a cumulative link model:*

$$P_\xi(X_j \leq \ell | \mathbf{X}_{\text{pa}(j)}, \mathbf{Z}) = F\left(\gamma_{j\ell} - \sum_{k \in \text{pa}(j)} \beta_{jk} X_k - \boldsymbol{\delta}_j' \mathbf{Z} - \alpha_j\right), \quad \ell = 1, \dots, L_j,$$

where F is a known, strictly increasing link function, $\gamma_{j1} < \dots < \gamma_{j,L_j-1} < \gamma_{j,L_j} = \infty$ are threshold parameters with $\gamma_{j1} = 0$, β_{jk} are parent effect parameters with $\beta_{jk,1} = 0$, $\boldsymbol{\delta}_j$ are covariate adjustment parameters, and α_j is an intercept. We require $L_j \geq 3$ for all $j = 1, \dots, q$.

REMARK 2. *The restriction $L_j \geq 3$ is necessary: if $L_j = 2$, the cumulative link model collapses to a standard binary regression and causal directionality is identifiable only up to the Markov equivalence class (Ni and Mallick, 2022).*

ASSUMPTION 5 (Ordinal Identifiability). *Let $\mathcal{G}_1$ and $\mathcal{G}_2$ be any two Markov-equivalent DAGs on $\mathbf{X}$, and let $\boldsymbol{\theta}_1$ denote a parameter under $\mathcal{G}_1$ in the no-covariate oBn (Assumption 4 with $\boldsymbol{\delta}_j = \mathbf{0}$ for all j). Define the pseudo-true parameter $\boldsymbol{\theta}_2^\dagger(\boldsymbol{\theta}_1) = \arg \max_{\boldsymbol{\theta}} \sum_{\mathbf{x}} P(\mathbf{x} | \mathcal{G}_1, \boldsymbol{\theta}_1) \log P(\mathbf{x} | \mathcal{G}_2, \boldsymbol{\theta})$. Then the set of $\boldsymbol{\theta}_1$ for which $P(\mathbf{X} | \mathcal{G}_1, \boldsymbol{\theta}_1) \equiv P(\mathbf{X} | \mathcal{G}_2, \boldsymbol{\theta}_2^\dagger(\boldsymbol{\theta}_1))$ is a proper analytic subvariety of $\Theta_{\mathcal{G}_1}$ of Lebesgue measure zero. We assume the true no-covariate parameter $\boldsymbol{\theta}^*$ lies outside this exceptional set.*

Assumption 5 is the central identifiability result of Ni and Mallick (2022) for the bivariate oBn ; its extension to multivariate DAGs follows from the pairwise decomposition argument of Ni, Chen and Wang (2025). Proposition 1 lifts this generic identifiability to the covariate-adjusted model.

ASSUMPTION 6 (Non-Degeneracy). *$P_\xi(X_j = \ell | \mathbf{X}_{\text{pa}(j)} = \mathbf{x}_{\text{pa}(j)}, \mathbf{Z} = \mathbf{z}) > 0$ for all j , all levels ℓ , all parent configurations $\mathbf{x}_{\text{pa}(j)}$, and all $\mathbf{z}$ in the support of $\mathbf{Z}$.*

ASSUMPTION 7 (Positivity). *Every unit $i \in U_N$ has a known, strictly positive first-order inclusion probability: $\pi_i = \Pr(i \in S_n) > 0$.*

ASSUMPTION 8 (Bounded Weights). *There exist constants $0 < c \leq C < \infty$, independent of N , such that $c \leq nw_i/N \leq C$ for all $i \in U_N$, ensuring that no single unit dominates the pseudo-likelihood.*

ASSUMPTION 9 (Design Consistency). *For any bounded, measurable function g on the joint sample space of $(\mathbf{X}, \mathbf{Z})$,*

$$\frac{1}{N} \sum_{i \in S_n} w_i g(\mathbf{x}_i, \mathbf{z}_i) - \frac{1}{N} \sum_{i \in U_N} g(\mathbf{x}_i, \mathbf{z}_i) \xrightarrow{p} 0$$

under the design distribution p_N , for each fixed realization of U_N .

Assumption 9 is satisfied by standard probability sampling designs under Assumptions 7–8; see Fuller (2009).

ASSUMPTION 10 (Regularity). *For each DAG $\mathcal{G}$: (i) the parameter space $\Theta_{\mathcal{G}}$ is compact; (ii) $\log P(\mathbf{x} \mid \mathcal{G}, \boldsymbol{\theta}, \mathbf{z})$ is twice continuously differentiable in $\boldsymbol{\theta}$ for each $(\mathbf{x}, \mathbf{z})$; (iii) the superpopulation expected log-likelihood $Q_{\xi}(\boldsymbol{\theta} \mid \mathcal{G}) = \mathbb{E}_{\xi}[\log P(\mathbf{X} \mid \mathcal{G}, \boldsymbol{\theta}, \mathbf{Z})]$ has a unique maximizer $\boldsymbol{\theta}_{\mathcal{G}}^*$ with nonsingular Hessian; and (iv) $\boldsymbol{\theta} \mapsto \log P(\mathbf{x} \mid \mathcal{G}, \boldsymbol{\theta}, \mathbf{z})$ satisfies a Lipschitz condition uniformly over $(\mathbf{x}, \mathbf{z})$.*

For the $\circ\text{Bn}$ with finitely many ordinal levels, condition (iv) is automatic since $\mathbf{X}$ has finite support and the link function F is smooth.

5.2. Main Results. Define the survey-weighted pseudo-log-likelihood under DAG $\mathcal{G}$ as $\tilde{\ell}(\boldsymbol{\theta} \mid \mathcal{G}) = \sum_{i \in S_n} w_i \log P(\mathbf{x}_i \mid \mathcal{G}, \boldsymbol{\theta}, \mathbf{z}_i)$, the pseudo-MLE as $\hat{\boldsymbol{\theta}}_{\mathcal{G}}^w = \arg \max_{\boldsymbol{\theta}} \tilde{\ell}(\boldsymbol{\theta} \mid \mathcal{G})$, and the survey-weighted pseudo-BIC as $\widetilde{\text{pBIC}}(\mathcal{G}) = -2\tilde{\ell}(\hat{\boldsymbol{\theta}}_{\mathcal{G}}^w \mid \mathcal{G}) + K_{\mathcal{G}} \cdot \rho_n$, where $K_{\mathcal{G}} = \sum_{j=1}^q (L_j - 1 + \sum_{k \in \text{pa}(j)} (L_k - 1) + \dim(\boldsymbol{\delta}_j))$ is the number of free parameters and ρ_n is a penalty sequence satisfying $\rho_n \rightarrow \infty$ and $\rho_n/N \rightarrow 0$. In practice, $\rho_n = \log(n_{\text{eff}})$.

REMARK 3 (Relation to the theoretical pseudo-BIC). *The pseudo-BIC (6) uses normalized weights ($\sum_{i \in S} w_i = n$) with likelihood rescaling n_{eff}/n and penalty $\log(n_{\text{eff}})$; this section analyzes the equivalent formulation with raw weights ($w_i = 1/\pi_i$) and penalty ρ_n . The ratio N/n_{eff} is graph-invariant, so $\arg \min_{\mathcal{G}} \widetilde{\text{pBIC}}(\mathcal{G})$ coincides under either parameterization.*

5.3. Covariate adjustment.

PROPOSITION 1 (Identifiability under covariate adjustment). *Suppose Assumption 5 holds for the no-covariate $\circ\text{Bn}$. Let $\mathcal{G}^*$ and $\mathcal{G}'$ be Markov-equivalent DAGs with $\mathcal{G}' \neq \mathcal{G}^*$. Then the set of $(\boldsymbol{\theta}^*, \boldsymbol{\delta}^*) \in \Theta_{\mathcal{G}^*} \times \mathbb{R}^{r^q}$ for which*

$$\mathbb{E}_{\mathbf{Z}} \left[\text{KL} \left(P_{\xi}(\mathbf{X} \mid \mathcal{G}^*, \boldsymbol{\theta}^*, \boldsymbol{\delta}^*, \mathbf{Z}) \parallel P(\mathbf{X} \mid \mathcal{G}', \boldsymbol{\theta}^{\dagger}(\mathbf{Z}), \mathbf{Z}) \right) \right] = 0$$

has Lebesgue measure zero. We assume throughout that the true parameters $(\boldsymbol{\theta}^, \boldsymbol{\delta}^*)$ of the data-generating model lie outside this exceptional set, in which case the expected KL divergence is strictly positive.*

PROPOSITION 2 (Pseudo-MLE Consistency). *Under Assumptions 4 and 6–10, for each DAG $\mathcal{G}$, $\hat{\boldsymbol{\theta}}_{\mathcal{G}}^w \xrightarrow{p} \boldsymbol{\theta}_{\mathcal{G}}^*$ under the joint $(\xi \times p_N)$ -probability, where $\boldsymbol{\theta}_{\mathcal{G}}^* = \arg \max_{\boldsymbol{\theta}} Q_{\xi}(\boldsymbol{\theta} \mid \mathcal{G})$.*

The proof of Proposition 2 proceeds in two stages, design consistency followed by superpopulation convergence (Binder, 1983; Fuller, 2009).

THEOREM 5.1 (Skeleton Consistency). *Under Assumptions 1–10, let $\mathcal{G}^*$ be the true DAG and $\mathcal{G}'$ any DAG not Markov-equivalent to $\mathcal{G}^*$. Then $\Pr \left(\widetilde{\text{pBIC}}(\mathcal{G}^*) < \widetilde{\text{pBIC}}(\mathcal{G}') \right) \rightarrow 1$.*

Theorem 5.1 validates the addition and deletion moves under either scoring variant: the penalized pseudo-BIC correctly recovers the skeleton of $\mathcal{G}^*$. Within Markov equivalence classes, where the skeleton is fixed, orientation must be settled by a different mechanism: under heterogeneous $\mathbf{L}$, single-BIC’s penalty terms do not cancel and introduce a directional bias toward parents with fewer ordinal levels (Section 4.1). Split-BIC eliminates this bias by basing reversal moves on the unpenalized pseudo-likelihood. The next theorem shows this comparison is correct in the limit.

THEOREM 5.2 (Orientation Consistency). *Under Assumptions 1–10, let $\mathcal{G}^*$ be the true DAG and $\mathcal{G}'$ any DAG Markov-equivalent to $\mathcal{G}^*$ with $\mathcal{G}' \neq \mathcal{G}^*$. Then we have $\Pr(\tilde{\ell}(\hat{\theta}_{\mathcal{G}^*}^w | \mathcal{G}^*) > \tilde{\ell}(\hat{\theta}_{\mathcal{G}'}^w | \mathcal{G}')) \rightarrow 1$.*

Theorem 5.2 compares the unpenalized pseudo-likelihood under $\mathcal{G}^*$ to that under any Markov-equivalent $\mathcal{G}'$ and shows the correctly oriented DAG wins as $N \rightarrow \infty$. Split-BIC scores reversal moves by exactly this unpenalized pseudo-likelihood, so split-BIC’s reversal decisions converge to the correct orientation.

Single-BIC scores reversal moves by the penalized pseudo-BIC, which adds a term $(L_k - L_j)\rho_n$ to the comparison between $k \rightarrow j$ and $j \rightarrow k$. The level difference $|L_k - L_j|$ is fixed by the data and does not grow with N , so this added term is $O(\rho_n)$. Since $\rho_n/N \rightarrow 0$ by assumption, it is asymptotically dominated by the $O(N)$ likelihood gap, and single-BIC’s reversal decisions also converge to the correct orientation. Both variants therefore inherit orientation consistency.

COROLLARY 1 (Full DAG Recovery). *Under Assumptions 1–10, the `swa-OBn` algorithm under either scoring variant recovers the true DAG $\mathcal{G}^*$ with probability tending to one.*

6. Application: Causal Architecture of Veteran Social Determinants of Health. We applied the `swa-OBn` to the VHA survey data of Section 2 to address the two scientific questions posed therein. Sensitivity analyses concerning the proportional odds working assumption are deferred to the Supplementary Material.

6.1. Estimated DAGs, bootstrap uncertainty, weighted versus unweighted comparisons. We ran `swa-OBn` with split-BIC scoring, $R = 10$ random restarts, the covariates $\mathbf{Z}$ of Section 2.4, and the outcome-to-need blacklist. Kish’s effective sample size was $n_{\text{eff}} = 1,387$ (design effect 3.89). Uncertainty was quantified by $B = 250$ bootstrap resamples (Section 4.4).

Figure 1 displays the bootstrap median-probability-model DAG for the weighted analysis (panel (a)) and the unweighted analysis (panel (b)). The weighted DAG retains 14 edges (8 robust, 6 suggestive); the unweighted DAG retains 23. Eleven node pairs are connected in both analyses, though for 3 of these pairs the inferred causal direction reverses between the weighted and unweighted DAGs — that is, a relationship estimated as $A \rightarrow B$ in one analysis is estimated as $B \rightarrow A$ in the other, indicating a different conclusion about which node causally precedes the other. Nine additional unweighted edges exist beyond the 14 in the weighted DAG. Childcare is isolated under weighting but receives two suggestive in-edges without it. The mental-health–well-being orientation is suggestive under weighting ($\hat{p}_{\text{WB} \rightarrow \text{MH}} = 0.54$) but is robust without weighting ($\hat{p}_{\text{WB} \rightarrow \text{MH}} = 0.85$). Full 10×10 inclusion matrices appear in Supplementary Table 5.

Despite a smaller effective penalty in the weighted score ($\log n_{\text{eff}} \approx 7.23$ vs. $\log n \approx 8.59$), the weighted DAG is the sparser of the two. The 9 unweighted-only edges therefore reflect associations whose log-likelihood signal survives the larger unweighted penalty but dissipates under reweighting — i.e., structure is driven by the oversampled strata (sampling-weight ratio $\max / \min \approx 980$). Two illustrations: isolation/loneliness/discrimination, whose out-degree drops from 8 to 4 with reversal of the basic-needs–isolation/loneliness/discrimination edge; and childcare, where borderline unweighted in-edges from caregiving ($\hat{p} = 0.54$) and work/education ($\hat{p} = 0.52$) vanish under reweighting. The 3 shared-pair orientation reversals arise through the same channel: the `OBn` criterion scores directions by cumulative log-likelihood differences along the L -direction, which reweighting can flip. Key implications are deferred to Section 8.

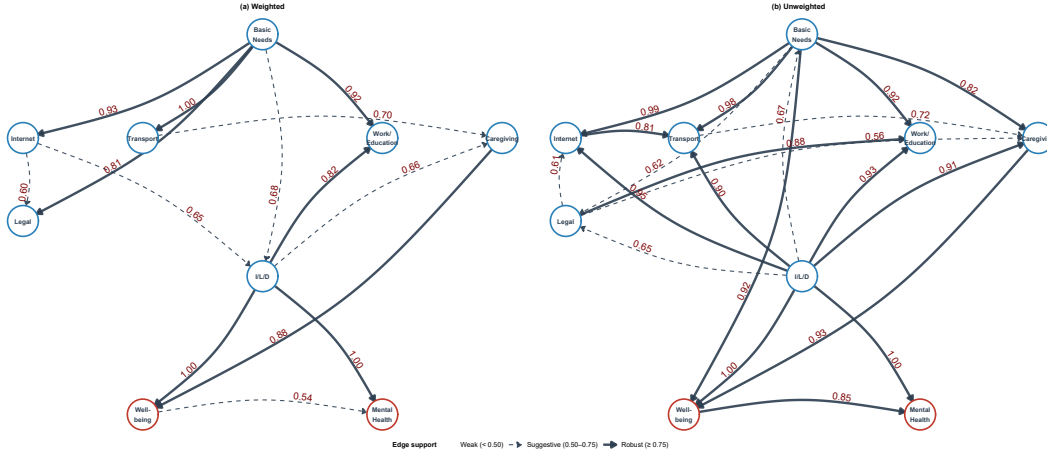


FIG 1. Consensus DAGs for veteran social needs and mental health outcomes, estimated by $swa-oBn$ on the VHA SHEP January–February 2023 primary care sample ($n = 5,396$), adjusted for age and race-ethnicity-sex. Panel (a) uses survey sampling weights; panel (b) sets all weights to one. Node positions are identical across panels; only edges differ. The self-reported social need nodes shown include: (1) basic needs (paying for basics like food, housing, medical care, and heating; paying for food (only); getting or maintaining housing); (2) work/education (finding or keeping work, getting additional education or job training); (3) isolation/loneliness/discrimination, abbreviated I/L/D (feeling socially isolated; feeling lonely; managing experiences of discrimination); (4) caregiving (adult caregiving for self or others); (5) childcare (obtaining childcare); (6) transport (transportation for basic needs like medical care or grocery shopping); (7) internet (getting internet at home); (8) legal (obtaining assistance for legal issues). Outcome-to-need edges are forbidden by blacklist. Edge probabilities are bootstrap inclusion frequencies over $B = 250$ resamples (algorithm converged at all 250 bootstrap samples); displayed edges have $\hat{p}_{jk} \geq 0.50$. Dashed edges are suggestive ($\hat{p} \in [0.50, 0.75]$), solid are robust ($\hat{p} \geq 0.75$). Childcare is isolated in (a) and receives suggestive in-edges from caregiving ($\hat{p} = 0.54$) and work/education ($\hat{p} = 0.52$) in (b).

6.2. *Key findings.* The weighted consensus DAG (Figure 1, panel (a)) answers the two scientific questions posed in Section 2: which needs are upstream root causes (finding 1), and which needs serve as *proximal* versus *distal* causes of the health indicators (findings 2–3).

1. *Root cause among needs:* Basic needs has out-edges to 5 of the 7 other connectable need nodes ($\hat{p}$ from 0.68 to 1.00) and no in-edges from any need node at $\hat{p} \geq 0.50$. No other need node has out-degree above 4.
2. *Proximal causes of the outcome nodes:* Three need-to-outcome edges have $\hat{p} \geq 0.50$: isolation, loneliness, and discrimination $\rightarrow$ mental health ($\hat{p} = 1.00$), isolation, loneliness, and discrimination $\rightarrow$ well-being ($\hat{p} = 1.00$), and caregiving $\rightarrow$ well-being ($\hat{p} = 0.88$). The two sources — the isolation, loneliness, and discrimination node and caregiving node — are therefore the *proximal* causes. The remaining six needs are *distal* and reach the health indicator nodes only through these three edges, except for childcare which is disconnected in the weighted DAG.
3. *Mental health has a unique proximal mediator:* Mental health receives a single direct in-edge from the need set: isolation, loneliness, and discrimination $\rightarrow$ mental health ($\hat{p} = 1.00$). No other need reaches the mental health indicator except through this mediator.

7. Simulation Studies. We evaluated the finite-sample performance of $swa-oBn$ through four simulation studies. Study 1 assessed the impact of survey design on DAG recovery; Study 2 examined covariate adjustment under confounding; Study 3 compared the single-BIC and split-BIC scoring variants of Section 4.1 across heterogeneous ordinal-level

configurations. Study 4 demonstrated that the survey-weighting problem extends to other structure-learning algorithm families (PC, FCI, LiNGAM), with full results in the Supplementary Material.

7.1. Common Setup. All simulations shared the following design. A finite population of $N = 20,000$ units was generated from an ordinal Bayesian network with $q = 8$ nodes, each having $L = 3$ ordinal levels except where noted. The true DAG was drawn once from an Erdős–Rényi model with edge probability 0.25 and held fixed across replications. Parent effects β_{jk} and intercepts α_j were drawn independently from $N(0, \sigma^2)$, where σ controls the signal strength of the cumulative link model (1).

From the population, a probability sample of approximate size n was drawn via Bernoulli sampling. Inclusion probabilities were assigned based on two designated oversampling variables (X_1 and X_q): each unit’s target score was $s_i = x_{i1} + x_{iq}$, and units in the top quartile of s_i received an elevated inclusion probability π_{hi} , with the rest receiving π_{lo} ($w_i = 1/\pi_i$). This two-stratum design induces informative sampling because the oversampling variables are nodes in the DAG. Three design tiers were considered, parameterized by the achieved design effect $\text{deff} = n/n_{\text{eff}}$: *none* (simple random sampling, $\text{deff} = 1$), *mild* ($\text{deff} \approx 1.5$), and *strong* ($\text{deff} \approx 3.2$). The strong tier was calibrated to approximate the design effect of the SHEP analytic sample of Section 6 ($\text{deff} = 3.89$).

Each scenario was replicated 100 times. Performance was measured by the structural Hamming distance (SHD) between the estimated and true DAGs, defined as the number of edge additions, deletions, or reversals required to transform one into the other.

7.2. Study 1: Effect of Survey Design on Structure Recovery. Study 1 compared $\text{swa-}\text{oBn}$ with the unweighted oBn of Ni, Chen and Wang (2025) through three sub-studies, each varying one factor (Figure 2). Panel (A) varied design strength at $n \approx 1,000$ and $\sigma = 1.5$. Edge precision and correct orientation rates are reported in the Supplementary Material. Under simple random sampling the methods coincided ($\text{SHD} = 2.94$ vs 2.93), confirming that subsequent differences were attributable to the weight correction. Under the mild design $\text{swa-}\text{oBn}$ reduced SHD from 4.08 to 3.28; under the strong design the corresponding values are 4.23 and 3.80. Panel (B) varied sample size under the strong design. Both methods improved with n , but the gap widened from 0.09 at $n = 300$ to 0.57 at $n = 4,000$. This pattern is the finite-sample signature of the bias characterized in Theorems 5.1 and 5.2: $\text{swa-}\text{oBn}$ converged toward the truth while oBn plateaued above it. Panel (C) varied signal strength. $\text{swa-}\text{oBn}$ achieved lower SHD at every value of $\sigma \in \{0.25, 0.75, 1.0, 1.5, 2.0\}$ and was statistically indistinguishable at $\sigma = 0.5$. The largest gap occurred at $\sigma = 1.5$ ($\text{SHD} = 4.16$ vs 3.70), where the ordinal identifiability mechanism retained sufficient signal to orient edges while design-induced bias remained strong enough to mislead the unweighted method.

7.3. Study 2: Value of Covariate Adjustment. Study 2 isolated the role of covariate adjustment and its interaction with survey weighting. The true DAG had $q = 8$ nodes partitioned into six upstream nodes and two terminal outcomes, with two parallel chains $X_1 \rightarrow X_2 \rightarrow X_3 \rightarrow X_4$ and $X_1 \rightarrow X_5 \rightarrow X_6$ emanating from a common root; outcome X_7 had parent X_3 , and outcome X_8 was a collider with parents X_4 and X_6 . A confounder $Z \sim N(0, 1)$ entered the cumulative link model (7) additively for nodes $\{X_2, X_3, X_5, X_7\}$ with common effect magnitude δ . The confounded set was disjoint from the oversampling variables $\{X_1, X_8\}$ so that survey-design and confounding biases were mechanically independent. Four estimators were compared: oBn (no weights, no Z), Z -adjusted (no weights, Z included), *weighted* (weights, no Z), and $\text{swa-}\text{oBn}$ (weights and Z). Panel (A) varied δ at the strong design with $n \approx 1,000$ and $\sigma = 1.5$. At $\delta = 0$ the four estimators were statistically indistinguishable. As δ grew, oBn and the weighted estimator degraded in parallel,

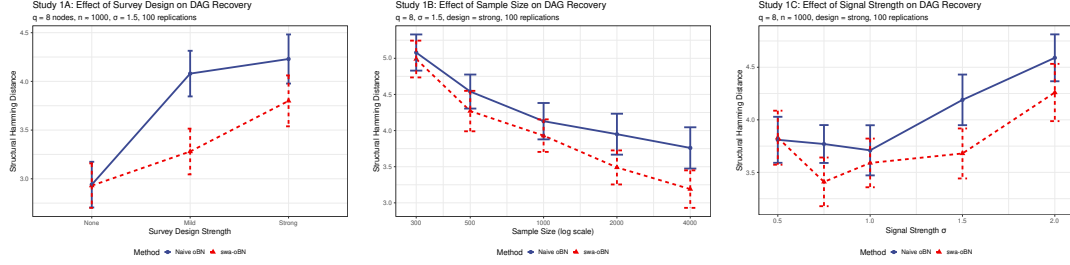


FIG 2. Mean SHD (± 1.96 SE) for $swa-oBn$ and oBn of *Ni, Chen and Wang (2025)* across 100 replications. (A) Design strength at $n \approx 1,000$, $\sigma = 1.5$. (B) Sample size under the strong design, $\sigma = 1.5$. (C) Signal strength under the strong design at $n \approx 1,000$. All panels: $q = 8$, $L = 3$.

reaching SHD = 7.59 and 7.33 at $\delta = 2.0$, confirming that weighting alone does not address confounding. The Z -adjusted and $swa-oBn$ estimators improved with δ , falling to 5.03 and 5.24 respectively. Panel (B) fixed $\delta = 1.2$ and varied the design. Under simple random sampling, Z -adjustment and $swa-oBn$ coincided (SHD = 4.10 each): weights were vacuous when sampling is uninformative. Under the mild design $swa-oBn$ attained the lowest SHD (4.57 vs 5.31 for Z -adjusted, 6.69 for weighted, 7.17 for naive). Under the strong design $swa-oBn$ remained best (5.77 vs 5.92 for Z -adjusted, 7.08 for weighted, 7.79 for naive), though the margin over Z -adjustment narrowed. Survey weighting and covariate adjustment thus address distinct bias sources, with their joint application offering the largest gains under intermediate design strength.

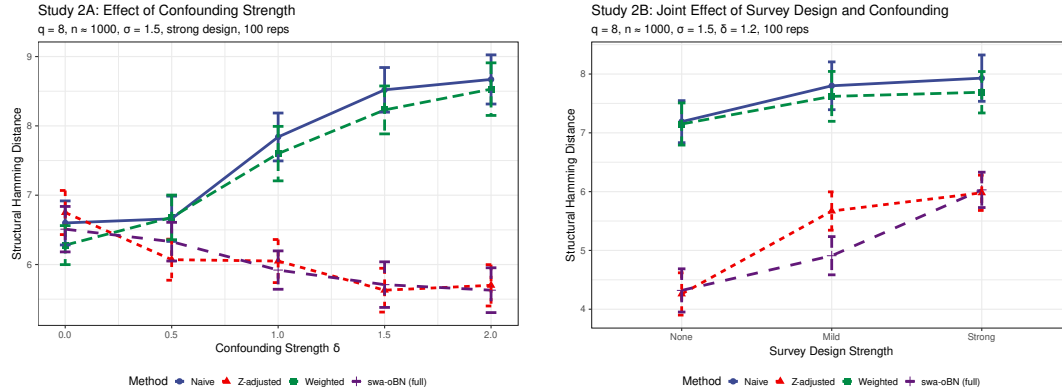


FIG 3. Mean SHD (± 1.96 SE) for four estimators across 100 replications. (A) Confounding strength δ under the strong design ($deff \approx 3.3$); weighting alone fails to address confounding, while Z -adjustment and $swa-oBn$ improve with δ . (B) Design strength with $\delta = 1.2$ fixed; $swa-oBn$ is best at every design tier and ties with Z -adjustment when sampling is uninformative. Both panels: $q = 8$, $\sigma = 1.5$.

7.4. Study 3: Single-BIC versus Split-BIC Scoring. Study 3 evaluated the two scoring variants of Section 4.1 across ordinal-level configurations. The true DAG had $q = 8$ nodes; the level vector $\mathbf{L} = (L_1, \dots, L_8)$ varied across five regimes (Table 1). Each regime was replicated 100 times at $n \in \{200, 400, 700, 1,000\}$ under the strong design, with four paired fits per replication: split-BIC and single-BIC, each with and without survey weighting. Table 1 reports the paired SHD gap (split-BIC – single-BIC) at $n = 1,000$ under survey weighting. Split-BIC outperformed single-BIC in every regime, with relative gains ranging from 1.7%

(Uniform) to 19.7% (Wide). The mechanism is that scoring reversals on the unpenalized pseudo-likelihood removed the $(L_k - L_j) \log n$ penalty differential of Section 4.1, leaving the orientation decision to the likelihood, where ordinal identifiability favors the correct direction. Supplementary Material confirmed the pattern was stable across n and across both survey-weighted and unweighted arms. Section 6 therefore uses split-BIC.

TABLE 1

Paired SHD gap (split-BIC – single-BIC) under survey weighting at $n = 1,000$. Negative values favor split-BIC. All gaps are significant at the $|\text{gap}| > 1.96 \cdot \text{SE}$ threshold.

Regime	L	Single-BIC	Split-BIC	Gap (SE)	Relative
Uniform	(3, 3, 3, 3, 3, 3, 3, 3)	4.12	4.05	−0.07 (0.02)	−1.7%
Adjacent	(3, 3, 3, 3, 3, 4, 4, 4)	4.54	4.08	−0.46 (0.06)	−10.2%
Polarized	(3, 3, 3, 3, 5, 5, 5, 5)	4.33	3.72	−0.61 (0.05)	−14.1%
Wide	(3, 3, 3, 3, 3, 3, 3, 7)	4.97	3.99	−0.98 (0.07)	−19.7%
Extreme	(3, 3, 3, 3, 3, 3, 3, 9)	4.87	4.06	−0.81 (0.06)	−16.6%

7.5. Study 4: Generality of the Survey-Weighting Problem. To confirm that the distortion introduced by informative sampling is not specific to ordinal Bayesian networks, we applied survey-weighted and unweighted versions of three additional algorithms: the PC algorithm, FCI, and LiNGAM, to a simple $X \rightarrow Y \rightarrow Z$ chain. Under mild and strong designs, the unweighted versions frequently infer a spurious X – Z edge, while their weighted counterparts recover the correct structure. Complete results appear in the Supplementary Material.

8. Discussion. This paper introduced the survey-weighted, covariate-adjusted ordinal Bayesian network (swa-OBn), a framework for causal discovery from complex survey data with ordinal responses. The method integrates sampling weights and Kish’s effective sample size into a pseudo-BIC scoring framework, develops a split-BIC scoring procedure that decouples skeleton control from edge orientation, supports exogenous covariate adjustment without expanding the DAG, and allows domain knowledge to be encoded via structural constraints. We proved that the framework consistently recovers the true population-level causal graph under a joint superpopulation-and-design asymptotic framework.

8.1. Key findings and implications. Applied to the VHA SHEP data, the swa-OBn revealed a causal architecture of Veteran social needs that could not have been discovered by the associational methods used in prior analyses of this dataset (Frank et al., 2025; Lamba et al., 2025; Russell et al., 2026). Three principal findings emerged. First, basic needs (food, housing, paying for basics) function as the upstream causal hub, with direct edges to five of the seven other need nodes. Second, isolation, loneliness, and discrimination serve as the proximal mediator of mental health and well-being, with bootstrap inclusion probabilities of 1.00 for both edges into the outcome layer. Third, mental health receives only a single direct in-edge from the need set — isolation, loneliness, and discrimination $\rightarrow$ mental health ($\hat{p} = 1.00$); all other social need-to-mental health information flows through this mediator or, secondarily, through caregiving and well-being. The structural placement of isolation, loneliness, and discrimination as a mediator rather than an upstream root is specifically a consequence of survey weighting. The survey design oversamples non-White-male Veteran subgroups; in those subgroups, prior associational analyses of these same data documented elevated isolation, loneliness, and discrimination-related needs alongside elevated

basic needs and reported need in other health-related social domains (Frank et al., 2025; Russell et al., 2026). The unweighted bootstrap reads the resulting sample-level co-occurrence as a common-cause pattern and re-orientes edges so that the isolation, loneliness, and discrimination node appears as a network root with direct influence on transportation, internet, legal assistance, and basic needs themselves. Survey weighting back to the superpopulation removes this structural distortion and recovers it as a mediator downstream of basic needs. The simulation evidence in Section 7.2 confirms that this is not an artifact of the survey data but a generic consequence of design-confounded structure that the `swa-oBn` corrects by construction.

The weighted DAG also clarifies an asymmetry in intervention prioritization. Resolving basic needs propagates to mental health primarily through isolation, loneliness, and discrimination, but this mediator has additional inputs that an intervention focused solely on addressing basic needs will be unlikely to reach—for example, social isolation related to rural geography, loneliness resulting from bereavement, and the systemic experience of discrimination. Conversely, interventions or programs targeting isolation, loneliness, or discrimination directly may produce short-term gains that erode or fail to result in long-term, sustainable change if upstream material deprivation is left unaddressed. The implication is therefore neither “treat basic needs and the mediator follows” nor “treat the mediator and the upstream cause is irrelevant”: meaningful improvement in mental health and well-being requires acting on both layers—the upstream roots and the downstream mediators—of the pathway, with neither layer sufficient alone.

The weighted DAG also bears directly on the design of social needs screening instruments currently deployed in VHA primary care and the broader healthcare system. Of the eight self-reported need domains in our analysis, only two carry direct edges into the mental-health-well-being outcome layer at $\hat{p} \geq 0.50$: isolation/loneliness/discrimination into both outcomes ($\hat{p} = 1.00$ each), and adult caregiving into well-being ($\hat{p} = 0.88$). Of the remaining six need domains, five—basic needs, work and education, transportation, internet, and legal assistance—reach the outcomes only indirectly, predominantly through the isolation/loneliness/discrimination mediator; childcare is disconnected from the outcome layer entirely in the weighted DAG. The caregiving edge is itself notable. Caregiver support is included in Boston Medical Center’s THRIVE screening and referral program (Boston Medical Center, 2026); however, many other commonly used screening instruments, including the CMS Accountable Health Communities screening tool (Billieux et al., 2017), Protocol for Responding to and Assessing Patients’ Assets, Risks, and Experiences (PRAPARE) screening (Weir et al., 2020), and the VHA Assessing Circumstances and Offering Resources for Needs (ACORN) social risk screening and follow up intervention (Russell et al., 2023) lack a measure to assess adult caregiving needs. Although timely access to care and benefits for caregivers is a stated VA organizational priority and the VHA has invested substantially in the Caregiver Support Program since its launch in 2011 (Miller et al., 2025), the VHA currently lacks a systematic way to identify Veterans with caregiving needs or the needs of Veteran caregivers. Given the findings from this study, future work should assess the feasibility of screening for caregiving needs as part of routine clinical care within VHA. Taken together, patterns observed in this Veteran sample suggest that broad health-related social domain screening instruments may include items whose information content for mental health and well-being is largely captured by their indirect operation through the psychosocial mediator, while at least one operationally consequential domain (i.e., caregiving) is absent from many current and commonly used instruments. Whether the breadth of routinely administered domains can be reduced without proportional loss of outcome-relevant signal is an open question worth comparative evaluation, particularly given that broad domain screening can impose a burden that impedes sustainability in primary-care settings (Griesemer et al., 2024).

Social isolation, loneliness, and the experience of discrimination are each well-documented public-health concerns (Schmitt et al., 2014; Grenawalt et al., 2022; Emmer, Dorn and Mata, 2024), and the elevated prevalence of all three in this Veteran sample is consistent with both disproportionate burden in subgroups oversampled in this survey (Hausmann et al., 2023; Russell et al., 2026) and the broader social isolation and loneliness trends documented since the COVID-19 pandemic (Hausmann et al., 2023). Direct intervention on these psychosocial domains is complicated: social isolation experienced by virtue of rural residence or loneliness following bereavement may not be amenable to short-term resource provision. Within the VHA, however, several pilot programs have shown promise in helping address social isolation or loneliness and promoting social connectedness, including peer-facilitated isolation interventions (Eliacin et al., 2023), Veteran Coffee Socials (Gorman et al., 2018), the Veteran Outreach Into the Community to Expand Social Support (VOICES) program (Gorman et al., 2022), and the Compassionate Contact Corps (Sullivan et al., 2021). Managing experiences of discrimination is likewise difficult to address through routine resource provision, though early evaluations of race- and ethnicity-based stress and trauma interventions are positive (Conway-Phillips et al., 2020; Saban et al., 2021; Taylor et al., 2024; Hall-Clark et al., 2024), and the inclusion of measures to assess discrimination experiences in routine social risk or social need screening instruments is itself an open operational question.

8.2. Limitations. The demographic covariates Z (age and the combined race-ethnicity-sex indicator) are the same variables adjusted for in all three companion analyses of this dataset, where they substantially altered prevalence ratios (Frank et al., 2025; Lamba et al., 2025) and relative risk ratios (Russell et al., 2026). Age captures cohort and life-stage effects; the race-ethnicity-sex composite absorbs the systematic disparities documented in those studies and reflects the stratified oversampling design. These are plausibly the dominant demographic confounders for the relationships among social need variables in this population. However, several variables that could confound need-need or need-outcome relationships — including rurality, married/partnered status, and household income — remain unmeasured in the survey data. We do not claim that causal sufficiency holds perfectly. Rather, we argue that the Z -adjusted DAG is more credible than the unadjusted one and that the methodology is correct conditional on the assumption. If richer covariate data become available in future survey administrations, they can be incorporated into the `swa-oBn` framework with no algorithmic changes; only the Z matrix needs to be expanded. The DAG-isolated status of the childcare node in the weighted analysis warrants a related caveat. In the unweighted DAG (Figure 1, panel b), childcare receives suggestive in-edges from caregiving ($\hat{p} = 0.54$) and work/education ($\hat{p} = 0.52$); these connections vanish under reweighting because childcare-need responses are concentrated in younger, oversampled subgroups (e.g., female Veterans) and become sparse once the analytic distribution shifts toward the older Veteran primary-care population. Thus, the isolated weighted finding in which childcare is not connected to other social needs or health indicators should not be generalized to younger Veteran or non-Veteran populations.

Several additional limitations merit discussion. First, the greedy hill-climbing search is guaranteed to find only a local optimum. The multi-start procedure with $R = 10$ random initializations mitigates this concern, and the bootstrap procedure implicitly explores the DAG space by re-running the search on resampled data; nevertheless, the estimated DAG is not guaranteed to be the global optimum. Second, the polychoric clustering used to construct composite nodes introduces a working-model approximation. The pointwise maximum across constituent three-level items, while still a three-level ordinal, does not in general satisfy an exact cumulative link specification conditional on its parents, so the identifiability guarantee of Ni and Mallick (2022) holds only approximately for these composite nodes.

We treat this as a deliberate trade-off: the alternative (entering all 15 raw items into the graph) produces an overly dense and uninterpretable structure. The bootstrap procedure provides some robustness against orientation artifacts arising from this approximation. Third, the global effective sample size n_{eff} used in the pseudo-BIC penalty does not capture node-specific design effects. Different variables may have different design effects depending on how strongly the oversampling correlates with each variable. Incorporating node-specific design effects would require estimating variable-level design effects and integrating them into the scoring, an extension we leave for future work. Fourth, the survey data are cross-sectional, so the discovered DAG represents a contemporaneous causal structure, not a temporal one. The blacklist encoding the assumption that outcomes do not cause needs is the minimal temporal constraint imposed; all other directionality is learned from the data. Fifth, the proportional odds structure of the cumulative link model is treated as an explicitly designated working assumption rather than a tested property of the data; the Supplementary Material documents diagnostic checks.

8.3. Extensions. The survey-weighting problem documented in this paper is not specific to ordinal Bayesian networks. The Supplementary Material demonstrates that the PC algorithm, FCI, and LiNGAM all suffer from analogous distortions when applied to complex survey data. Developing fully survey-weighted extensions of these algorithms for multivariate settings, with formal consistency guarantees comparable to those established here for the `swa-OBn`, is a natural next step. A natural sensitivity extension is to examine the recovered DAG under alternative groupings of the psychosocial items — for instance, entering discrimination as a standalone node rather than within the isolation, loneliness, and discrimination composite — to assess robustness of the structural findings to polychoric clustering choices. Finally, longitudinal extensions of the SHEP, if available, would enable temporal causal discovery and permit relaxation of the cross-sectional assumption that currently limits the interpretation of the discovered graph.

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SUPPLEMENTARY MATERIAL

Supplementary Material for “Survey-Weighted Ordinal Causal Discovery: Application to Veteran Social Needs and Health Indicators”

Full proofs of Propositions 1–2 and Theorems 5.1–5.2; the `swa-OBn` algorithm in pseudocode; polychoric clustering details; Study 1 supplementary metrics; Study 3 supplementary tables; Study 4 results (survey-weighted PC, FCI, LiNGAM); proportional odds diagnostics; survey weight distribution details; the composite node table; complete bootstrap edge probability matrix for weighted and unweighted models.

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